

# assessment

November 13, 2024

```
[94]: import pandas as pd
```

```
df = pd.read_csv('tsa_claims2.csv')
```

```
df
```

```
c:\Users\HOME-PC\anaconda3\lib\site-  
packages\IPython\core\interactiveshell.py:3444: DtypeWarning: Columns (0,11,12)  
have mixed types.Specify dtype option on import or set low_memory=False.  
exec(code_obj, self.user_global_ns, self.user_ns)
```

```
[94]:
```

|        | Claim Number  | Date Received | Incident Date   | Airport Code | \ |
|--------|---------------|---------------|-----------------|--------------|---|
| 0      | 0909802M      | 4-Jan-02      | 12/12/2002 0:00 | EWR          |   |
| 1      | 0202417M      | 2-Feb-02      | 1/16/2004 0:00  | SEA          |   |
| 2      | 0202445M      | 4-Feb-02      | 11/26/2003 0:00 | STL          |   |
| 3      | 0909816M      | 7-Feb-02      | 1/6/2003 0:00   | MIA          |   |
| 4      | 2005032379513 | 18-Feb-02     | 2/5/2005 0:00   | MCO          |   |
| ...    | ...           | ...           | ...             | ...          |   |
| 204262 | 2015120427297 | 20-Nov-15     | 16-Oct-15       | -            |   |
| 204263 | 2015123027969 | 17-Dec-15     | 2-Dec-15        | -            |   |
| 204264 | 2016010428072 | 22-Dec-15     | 20-Dec-15       | -            |   |
| 204265 | 2016011328300 | 30-Dec-15     | 28-Dec-15       | -            |   |
| 204266 | 2015123128015 | 31-Dec-15     | 23-Nov-15       | -            |   |

|        | Airport Name                    | Airline Name         | \ |
|--------|---------------------------------|----------------------|---|
| 0      | Newark International Airport    | Continental Airlines |   |
| 1      | Seattle-Tacoma International    | NaN                  |   |
| 2      | Lambert St. Louis International | American Airlines    |   |
| 3      | Miami International Airport     | American Airlines    |   |
| 4      | Orlando International Airport   | Delta (Song)         |   |
| ...    | ...                             | ...                  |   |
| 204262 | -                               | -                    |   |
| 204263 | -                               | -                    |   |
| 204264 | -                               | -                    |   |
| 204265 | -                               | -                    |   |
| 204266 | -                               | -                    |   |

|        | Claim Type              | Claim Site \    |
|--------|-------------------------|-----------------|
| 0      | Property Damage         | Checkpoint      |
| 1      | Property Damage         | Checked Baggage |
| 2      | Property Damage         | Checked Baggage |
| 3      | Property Damage         | Checkpoint      |
| 4      | Property Damage         | Checkpoint      |
| ...    | ...                     | ...             |
| 204262 | Property Damage         | Checked Baggage |
| 204263 | Property Damage         | Checked Baggage |
| 204264 | Passenger Property Loss | Checked Baggage |
| 204265 | Passenger Property Loss | Checked Baggage |
| 204266 | Passenger Property Loss | Checkpoint      |

|        | Item                                              | Claim Amount \ |
|--------|---------------------------------------------------|----------------|
| 0      | Other                                             | \$350.00       |
| 1      | Luggage (all types including footlockers)         | \$100.00       |
| 2      | Cell Phones                                       | \$278.88       |
| 3      | Luggage (all types including footlockers)         | \$50.00        |
| 4      | Baby - Strollers; car seats; playpen; etc.        | \$84.79        |
| ...    | ...                                               | ...            |
| 204262 | Baggage/Cases/Purses; Books; Magazines & Other... | -              |
| 204263 | Audio/Video; Home Decor                           | -              |
| 204264 | Clothing                                          | -              |
| 204265 | Tools & Home Improvement Supplies                 | -              |
| 204266 | Personal Accessories                              | -              |

|        | Status   | Close Amount | Disposition     |
|--------|----------|--------------|-----------------|
| 0      | Approved | \$350.00     | Approve in Full |
| 1      | Settled  | \$50.00      | Settle          |
| 2      | Settled  | \$227.92     | Settle          |
| 3      | Approved | \$50.00      | Approve in Full |
| 4      | Approved | \$84.79      | Approve in Full |
| ...    | ...      | ...          | ...             |
| 204262 | -        | NaN          | NaN             |
| 204263 | -        | NaN          | NaN             |
| 204264 | -        | NaN          | NaN             |
| 204265 | -        | NaN          | NaN             |
| 204266 | -        | NaN          | NaN             |

[204267 rows x 13 columns]

```
[23]: df.info( verbose=True,memory_usage='deep',show_counts=True)
```

```
#info method is used to get a concise summary of a DataFrame. It provides
↪ information about the DataFrame.
#verbose parameter ensures that the method returns info for all of the columns.
#memory_usage='deep' ensures that the memory usage information is accurate.
```

```
#show_counts=True ensures that the non-null counts for each column will be  
↪displayed.
```

```
<class 'pandas.core.frame.DataFrame'>  
RangeIndex: 204267 entries, 0 to 204266  
Data columns (total 13 columns):  
#   Column                Non-Null Count  Dtype  
---  -  
0   Claim Number          204267 non-null object  
1   Date Received         204004 non-null object  
2   Incident Date         202084 non-null object  
3   Airport Code          195743 non-null object  
4   Airport Name          195743 non-null object  
5   Airline Name          169893 non-null object  
6   Claim Type            196354 non-null object  
7   Claim Site            203527 non-null object  
8   Item                  200301 non-null object  
9   Claim Amount          200224 non-null object  
10  Status                204262 non-null object  
11  Close Amount          135315 non-null object  
12  Disposition           131359 non-null object  
dtypes: object(13)  
memory usage: 170.3 MB
```

```
[24]: df.nunique()  
  
#Returns the number of unique values in each column
```

```
[24]: Claim Number          204258  
Date Received            4007  
Incident Date           26807  
Airport Code             465  
Airport Name             485  
Airline Name             318  
Claim Type               11  
Claim Site               6  
Item                    4704  
Claim Amount            36117  
Status                  15  
Close Amount            17162  
Disposition              3  
dtype: int64
```

```
[41]: # df.apply(pd.unique)  
#df.apply(pd.unique) applies the pd.unique function to each column of the  
↪DataFrame df.
```

```
# This means that for each column, it returns an array containing the unique
↳ values present in that column
```

```
df.rename(columns={'Airport Code': 'AirportCode'}, inplace=True)
df.rename(columns={'Claim Number': 'Claim Number'}, inplace=True)
df.rename(columns={'Date Received': 'DateReceived'}, inplace=True)
df.rename(columns={'Incident Date': 'IncidentDate'}, inplace=True)
df.rename(columns={'Airport Name': 'AirportName'}, inplace=True)
df.rename(columns={'Airline Name': 'AirlineName'}, inplace=True)
df.rename(columns={'Claim Type': 'ClaimType'}, inplace=True)
df.rename(columns={'Claim Site': 'ClaimSite'}, inplace=True)
df.rename(columns={'Claim Amount': 'ClaimAmount'}, inplace=True)
df.rename(columns={'Close Amount': 'CloseAmount'}, inplace=True)

df
# df.query("`Airport Code` == "EWR"")
```

```
[41]:
```

|        | Claim Number  | DateReceived | IncidentDate    | AirportCode | \ |
|--------|---------------|--------------|-----------------|-------------|---|
| 0      | 0909802M      | 4-Jan-02     | 12/12/2002 0:00 | EWR         |   |
| 1      | 0202417M      | 2-Feb-02     | 1/16/2004 0:00  | SEA         |   |
| 2      | 0202445M      | 4-Feb-02     | 11/26/2003 0:00 | STL         |   |
| 3      | 0909816M      | 7-Feb-02     | 1/6/2003 0:00   | MIA         |   |
| 4      | 2005032379513 | 18-Feb-02    | 2/5/2005 0:00   | MCO         |   |
| ...    | ...           | ...          | ...             | ...         |   |
| 204262 | 2015120427297 | 20-Nov-15    | 16-Oct-15       | -           |   |
| 204263 | 2015123027969 | 17-Dec-15    | 2-Dec-15        | -           |   |
| 204264 | 2016010428072 | 22-Dec-15    | 20-Dec-15       | -           |   |
| 204265 | 2016011328300 | 30-Dec-15    | 28-Dec-15       | -           |   |
| 204266 | 2015123128015 | 31-Dec-15    | 23-Nov-15       | -           |   |

|        | AirportName                     | AirlineName          | \ |
|--------|---------------------------------|----------------------|---|
| 0      | Newark International Airport    | Continental Airlines |   |
| 1      | Seattle-Tacoma International    | NaN                  |   |
| 2      | Lambert St. Louis International | American Airlines    |   |
| 3      | Miami International Airport     | American Airlines    |   |
| 4      | Orlando International Airport   | Delta (Song)         |   |
| ...    | ...                             | ...                  |   |
| 204262 | -                               | -                    |   |
| 204263 | -                               | -                    |   |
| 204264 | -                               | -                    |   |
| 204265 | -                               | -                    |   |
| 204266 | -                               | -                    |   |

|   | ClaimType       | ClaimSite       | \ |
|---|-----------------|-----------------|---|
| 0 | Property Damage | Checkpoint      |   |
| 1 | Property Damage | Checked Baggage |   |
| 2 | Property Damage | Checked Baggage |   |

|        |                         |                 |
|--------|-------------------------|-----------------|
| 3      | Property Damage         | Checkpoint      |
| 4      | Property Damage         | Checkpoint      |
| ...    | ...                     | ...             |
| 204262 | Property Damage         | Checked Baggage |
| 204263 | Property Damage         | Checked Baggage |
| 204264 | Passenger Property Loss | Checked Baggage |
| 204265 | Passenger Property Loss | Checked Baggage |
| 204266 | Passenger Property Loss | Checkpoint      |

|        | Item                                              | ClaimAmount | \ |
|--------|---------------------------------------------------|-------------|---|
| 0      | Other                                             | \$350.00    |   |
| 1      | Luggage (all types including footlockers)         | \$100.00    |   |
| 2      | Cell Phones                                       | \$278.88    |   |
| 3      | Luggage (all types including footlockers)         | \$50.00     |   |
| 4      | Baby - Strollers; car seats; playpen; etc.        | \$84.79     |   |
| ...    | ...                                               | ...         |   |
| 204262 | Baggage/Cases/Purses; Books; Magazines & Other... | -           |   |
| 204263 | Audio/Video; Home Decor                           | -           |   |
| 204264 | Clothing                                          | -           |   |
| 204265 | Tools & Home Improvement Supplies                 | -           |   |
| 204266 | Personal Accessories                              | -           |   |

|        | Status   | CloseAmount | Disposition     |
|--------|----------|-------------|-----------------|
| 0      | Approved | \$350.00    | Approve in Full |
| 1      | Settled  | \$50.00     | Settle          |
| 2      | Settled  | \$227.92    | Settle          |
| 3      | Approved | \$50.00     | Approve in Full |
| 4      | Approved | \$84.79     | Approve in Full |
| ...    | ...      | ...         | ...             |
| 204262 | -        | NaN         | NaN             |
| 204263 | -        | NaN         | NaN             |
| 204264 | -        | NaN         | NaN             |
| 204265 | -        | NaN         | NaN             |
| 204266 | -        | NaN         | NaN             |

[204266 rows x 13 columns]

```
[26]: df['ClaimType'].value_counts()

#Value_counts returns the count for each of the unique values in the Claim Type
↳column
```

```
[26]: Passenger Property Loss    117868
Property Damage                75364
Personal Injury                1465
Employee Loss (MPCECA)         485
Passenger Theft                 465
```

|                |     |
|----------------|-----|
| Motor Vehicle  | 369 |
| -              | 282 |
| Complaint      | 48  |
| Wrongful Death | 4   |
| Compliment     | 3   |
| Bus Terminal   | 1   |

Name: ClaimType, dtype: int64

```
[27]: df[['ClaimType', 'ClaimAmount']].value_counts()
#
```

```
[27]: ClaimType      ClaimAmount
Passenger Property Loss  $0.00      18775
Property Damage          $0.00      9350
Passenger Property Loss  -         7864
Property Damage          -         4206
                        $100.00     2077
                        ...
Passenger Property Loss  $618.26      1
                        $618.15      1
                        $1;459.35    1
                        $617.98      1
Wrongful Death          -           1
Length: 43430, dtype: int64
```

```
[28]: df[['ClaimType', 'AirportName']].value_counts()
```

```
[28]: ClaimType      AirportName
Passenger Property Loss  John F. Kennedy International      6334
                        Los Angeles International Airport    4320
                        Newark International Airport          4302
                        Orlando International Airport          4051
                        Miami International Airport            3937
                        ...
                        Ogdensburg International              1
                        Naples Municipal Airport               1
                        Molokai Airport                        1
                        Mckellar-Sipes Regional               1
Wrongful Death          Non TSA Airport (motor vehicle)    1
Length: 1475, dtype: int64
```

```
[29]: df[['ClaimType', 'ClaimSite']].value_counts()
```

```
[29]: ClaimType      ClaimSite
Passenger Property Loss  Checked Baggage      99708
Property Damage          Checked Baggage      54083
                        Checkpoint             20572
```

|                         |                 |       |
|-------------------------|-----------------|-------|
| Passenger Property Loss | Checkpoint      | 17440 |
| Personal Injury         | Checkpoint      | 974   |
| Property Damage         | Other           | 477   |
| Passenger Property Loss | Other           | 434   |
| Passenger Theft         | Checked Baggage | 396   |
| Employee Loss (MPCECA)  | Checked Baggage | 383   |
| Motor Vehicle           | Motor Vehicle   | 320   |
| Personal Injury         | Checked Baggage | 248   |
|                         | Other           | 151   |
| -                       | -               | 127   |
|                         | Checked Baggage | 99    |
| Passenger Property Loss | -               | 92    |
| Personal Injury         | Motor Vehicle   | 70    |
| Employee Loss (MPCECA)  | Checkpoint      | 66    |
| Passenger Theft         | Checkpoint      | 66    |
| -                       | Checkpoint      | 49    |
| Property Damage         | Motor Vehicle   | 49    |
|                         | -               | 40    |
| Motor Vehicle           | Checked Baggage | 26    |
| Complaint               | Checkpoint      | 21    |
|                         | Checked Baggage | 20    |
| Employee Loss (MPCECA)  | Motor Vehicle   | 19    |
|                         | Other           | 15    |
| Motor Vehicle           | Other           | 10    |
| Personal Injury         | -               | 10    |
| Passenger Property Loss | Bus Station     | 9     |
| Motor Vehicle           | Checkpoint      | 9     |
| Property Damage         | Bus Station     | 7     |
| Passenger Property Loss | Motor Vehicle   | 6     |
| -                       | Other           | 5     |
| Complaint               | Other           | 4     |
| Wrongful Death          | Other           | 3     |
| Complaint               | -               | 3     |
| Motor Vehicle           | -               | 3     |
| Compliment              | Checkpoint      | 2     |
| -                       | Motor Vehicle   | 2     |
| Bus Terminal            | Checked Baggage | 1     |
| Passenger Theft         | Other           | 1     |
| Compliment              | Checked Baggage | 1     |
| Employee Loss (MPCECA)  | -               | 1     |
| Wrongful Death          | Motor Vehicle   | 1     |
| Passenger Theft         | Motor Vehicle   | 1     |

dtype: int64

```
[30]: df[['ClaimType', 'Item']].value_counts()
```

```
[30]: ClaimType      Item
      Passenger Property Loss  Other
      21193
      Property Damage      Luggage (all types including footlockers)
      16814
                                   Other
      12740
      Passenger Property Loss  Clothing - Shoes; belts; accessories; etc.
      9438
                                   Jewelry - Fine
      8475
                                   ...
                                   Cosmetics - Perfume; toilet articles; medicines; soaps;
etc.; Currency; Jewelry - Fine; Other; PDA - Personal Data Assistants
      1
                                   Cosmetics - Perfume; toilet articles; medicines; soaps;
etc.; Currency; Jewelry - Fine; Paintings and Pictures (including frames;
etchings; photos; etc)      1
                                   Cosmetics - Perfume; toilet articles; medicines; soaps;
etc.; Currency; Medicines
      1
                                   Cosmetics - Perfume; toilet articles; medicines; soaps;
etc.; Currency; Musical Instruments - Other - inexpensive $100 or less
      1
                                   Clothing; Jewelry & Watches; Jewelry & Watches; Jewelry
& Watches
      1
Length: 5632, dtype: int64
```

```
[31]: df[df.duplicated(keep=False)]
```

```
[31]:      Claim Number Date Received      IncidentDate AirportCode AirportName \
16841  2004050450432    26-Sep-03  9/13/2003 0:00      NaN      NaN
16842  2004050450432    26-Sep-03  9/13/2003 0:00      NaN      NaN

      AirlineName      ClaimType      ClaimSite      Item \
16841      NaN  Property Damage  Checked Baggage  DVD/CD Players
16842      NaN  Property Damage  Checked Baggage  DVD/CD Players

      ClaimAmount      Status CloseAmount Disposition
16841    $31.96  Canceled      NaN      NaN
16842    $31.96  Canceled      NaN      NaN
```

```
[32]: df.drop_duplicates(keep='first', inplace=True)
```

```
df
```



[32]:

|        | Claim Number  | Date Received | IncidentDate    | AirportCode | \ |
|--------|---------------|---------------|-----------------|-------------|---|
| 0      | 0909802M      | 4-Jan-02      | 12/12/2002 0:00 | EWR         |   |
| 1      | 0202417M      | 2-Feb-02      | 1/16/2004 0:00  | SEA         |   |
| 2      | 0202445M      | 4-Feb-02      | 11/26/2003 0:00 | STL         |   |
| 3      | 0909816M      | 7-Feb-02      | 1/6/2003 0:00   | MIA         |   |
| 4      | 2005032379513 | 18-Feb-02     | 2/5/2005 0:00   | MCO         |   |
| ...    | ...           | ...           | ...             | ...         |   |
| 204262 | 2015120427297 | 20-Nov-15     | 16-Oct-15       | -           |   |
| 204263 | 2015123027969 | 17-Dec-15     | 2-Dec-15        | -           |   |
| 204264 | 2016010428072 | 22-Dec-15     | 20-Dec-15       | -           |   |
| 204265 | 2016011328300 | 30-Dec-15     | 28-Dec-15       | -           |   |
| 204266 | 2015123128015 | 31-Dec-15     | 23-Nov-15       | -           |   |

|        | AirportName                     | AirlineName          | \ |
|--------|---------------------------------|----------------------|---|
| 0      | Newark International Airport    | Continental Airlines |   |
| 1      | Seattle-Tacoma International    | NaN                  |   |
| 2      | Lambert St. Louis International | American Airlines    |   |
| 3      | Miami International Airport     | American Airlines    |   |
| 4      | Orlando International Airport   | Delta (Song)         |   |
| ...    | ...                             | ...                  |   |
| 204262 | -                               | -                    |   |
| 204263 | -                               | -                    |   |
| 204264 | -                               | -                    |   |
| 204265 | -                               | -                    |   |
| 204266 | -                               | -                    |   |

|        | ClaimType               | ClaimSite       | \ |
|--------|-------------------------|-----------------|---|
| 0      | Property Damage         | Checkpoint      |   |
| 1      | Property Damage         | Checked Baggage |   |
| 2      | Property Damage         | Checked Baggage |   |
| 3      | Property Damage         | Checkpoint      |   |
| 4      | Property Damage         | Checkpoint      |   |
| ...    | ...                     | ...             |   |
| 204262 | Property Damage         | Checked Baggage |   |
| 204263 | Property Damage         | Checked Baggage |   |
| 204264 | Passenger Property Loss | Checked Baggage |   |
| 204265 | Passenger Property Loss | Checked Baggage |   |
| 204266 | Passenger Property Loss | Checkpoint      |   |

|        | Item                                              | ClaimAmount | \ |
|--------|---------------------------------------------------|-------------|---|
| 0      | Other                                             | \$350.00    |   |
| 1      | Luggage (all types including footlockers)         | \$100.00    |   |
| 2      | Cell Phones                                       | \$278.88    |   |
| 3      | Luggage (all types including footlockers)         | \$50.00     |   |
| 4      | Baby - Strollers; car seats; playpen; etc.        | \$84.79     |   |
| ...    | ...                                               | ...         |   |
| 204262 | Baggage/Cases/Purses; Books; Magazines & Other... | -           |   |

|        |                                   |   |
|--------|-----------------------------------|---|
| 204263 | Audio/Video; Home Decor           | - |
| 204264 | Clothing                          | - |
| 204265 | Tools & Home Improvement Supplies | - |
| 204266 | Personal Accessories              | - |

|        | Status   | CloseAmount | Disposition     |
|--------|----------|-------------|-----------------|
| 0      | Approved | \$350.00    | Approve in Full |
| 1      | Settled  | \$50.00     | Settle          |
| 2      | Settled  | \$227.92    | Settle          |
| 3      | Approved | \$50.00     | Approve in Full |
| 4      | Approved | \$84.79     | Approve in Full |
| ...    | ...      | ...         | ...             |
| 204262 | -        | NaN         | NaN             |
| 204263 | -        | NaN         | NaN             |
| 204264 | -        | NaN         | NaN             |
| 204265 | -        | NaN         | NaN             |
| 204266 | -        | NaN         | NaN             |

[204266 rows x 13 columns]

```
[56]: newdf = df.query("`DateReceived` >= `IncidentDate`")
```

```
#Making Sure Data Is
newdf
```

```
[56]:
```

|        | Claim Number  | DateReceived | IncidentDate    | AirportCode | \ |
|--------|---------------|--------------|-----------------|-------------|---|
| 0      | 0909802M      | 4-Jan-02     | 12/12/2002 0:00 | EWR         |   |
| 1      | 0202417M      | 2-Feb-02     | 1/16/2004 0:00  | SEA         |   |
| 2      | 0202445M      | 4-Feb-02     | 11/26/2003 0:00 | STL         |   |
| 3      | 0909816M      | 7-Feb-02     | 1/6/2003 0:00   | MIA         |   |
| 11     | 2005070788431 | 7-Jul-02     | 6/17/2005 0:00  | CLT         |   |
| ...    | ...           | ...          | ...             | ...         |   |
| 204261 | 2015121427528 | 17-Nov-15    | 1-Nov-15        | -           |   |
| 204262 | 2015120427297 | 20-Nov-15    | 16-Oct-15       | -           |   |
| 204264 | 2016010428072 | 22-Dec-15    | 20-Dec-15       | -           |   |
| 204265 | 2016011328300 | 30-Dec-15    | 28-Dec-15       | -           |   |
| 204266 | 2015123128015 | 31-Dec-15    | 23-Nov-15       | -           |   |

|        | AirportName                             | AirlineName          | \ |
|--------|-----------------------------------------|----------------------|---|
| 0      | Newark International Airport            | Continental Airlines |   |
| 1      | Seattle-Tacoma International            | NaN                  |   |
| 2      | Lambert St. Louis International         | American Airlines    |   |
| 3      | Miami International Airport             | American Airlines    |   |
| 11     | Charlotte/Douglas International Airport | AirTran Airlines     |   |
| ...    | ...                                     | ...                  |   |
| 204261 | -                                       | -                    |   |
| 204262 | -                                       | -                    |   |

|        |   |   |
|--------|---|---|
| 204264 | - | - |
| 204265 | - | - |
| 204266 | - | - |

|        | ClaimType               | ClaimSite \     |
|--------|-------------------------|-----------------|
| 0      | Property Damage         | Checkpoint      |
| 1      | Property Damage         | Checked Baggage |
| 2      | Property Damage         | Checked Baggage |
| 3      | Property Damage         | Checkpoint      |
| 11     | Property Damage         | Checked Baggage |
| ...    | ...                     | ...             |
| 204261 | Passenger Property Loss | Checked Baggage |
| 204262 | Property Damage         | Checked Baggage |
| 204264 | Passenger Property Loss | Checked Baggage |
| 204265 | Passenger Property Loss | Checked Baggage |
| 204266 | Passenger Property Loss | Checkpoint      |

|        | Item                                              | ClaimAmount \ |
|--------|---------------------------------------------------|---------------|
| 0      | Other                                             | \$350.00      |
| 1      | Luggage (all types including footlockers)         | \$100.00      |
| 2      | Cell Phones                                       | \$278.88      |
| 3      | Luggage (all types including footlockers)         | \$50.00       |
| 11     | Luggage (all types including footlockers)         | \$43.49       |
| ...    | ...                                               | ...           |
| 204261 | Clothing; Household Items                         | -             |
| 204262 | Baggage/Cases/Purses; Books; Magazines & Other... | -             |
| 204264 | Clothing                                          | -             |
| 204265 | Tools & Home Improvement Supplies                 | -             |
| 204266 | Personal Accessories                              | -             |

|        | Status   | CloseAmount | Disposition     |
|--------|----------|-------------|-----------------|
| 0      | Approved | \$350.00    | Approve in Full |
| 1      | Settled  | \$50.00     | Settle          |
| 2      | Settled  | \$227.92    | Settle          |
| 3      | Approved | \$50.00     | Approve in Full |
| 11     | Approved | \$43.49     | Approve in Full |
| ...    | ...      | ...         | ...             |
| 204261 | -        | NaN         | NaN             |
| 204262 | -        | NaN         | NaN             |
| 204264 | -        | NaN         | NaN             |
| 204265 | -        | NaN         | NaN             |
| 204266 | -        | NaN         | NaN             |

[86872 rows x 13 columns]

Q1

```
[50]: common_claim = newdf['ClaimType'].value_counts().idxmax()

common_claim
```

```
[50]: 'Passenger Property Loss'
```

Q2

```
[51]: common_claimSite = newdf['ClaimSite'].value_counts().idxmax()

common_claimSite
```

```
[51]: 'Checked Baggage'
```

Q3

```
[58]: # What type of claim is made most at each claim site?
site_claim_counts = newdf.groupby(['ClaimSite', 'ClaimType']).size().
    ↪reset_index(name='Count')

most_common_claims = site_claim_counts.loc[site_claim_counts.
    ↪groupby('ClaimSite')['Count'].idxmax()]

most_common_claims
```

```
[58]:
```

|    | ClaimSite       | ClaimType               | Count |
|----|-----------------|-------------------------|-------|
| 0  | -               | -                       | 58    |
| 7  | Bus Station     | Passenger Property Loss | 6     |
| 14 | Checked Baggage | Passenger Property Loss | 41712 |
| 26 | Checkpoint      | Property Damage         | 9198  |
| 29 | Motor Vehicle   | Motor Vehicle           | 135   |
| 40 | Other           | Property Damage         | 215   |

Q4

```
[ ]: # What is the typical claim amount?
typicalClaimAmount = newdf.groupby('ClaimAmount').transform(func='mean')

typicalClaimAmount
```

Q5

```
[82]: # What is the overall claim approval rate for the entire U.S.?

total_claims = len(newdf)

# Count number of approved claims
```

```

approved_claims = newdf[newdf['Status'] == 'Approved'].shape[0]

# Calculate the approval rate
approval_rate = (approved_claims / total_claims) * 100

# Display the approval rate
approval_rate

```

[82]: 17.133253522423797

Q6

```

[138]: approved_or_settled = newdf[newdf['Status'].isin(['Approved', 'Settled'])]

newdf.loc[:, 'CloseAmount'] = newdf['CloseAmount'].astype(str).str.replace(r'[$;
↵-]', '', regex=True)
newdf.loc[:, 'ClaimAmount'] = newdf['ClaimAmount'].astype(str).str.replace(r'[$;
↵-]', '', regex=True)

newdf.loc[:, 'CloseAmount'] = pd.to_numeric(newdf['CloseAmount'],
↵errors='coerce')
newdf.loc[:, 'ClaimAmount'] = pd.to_numeric(newdf['ClaimAmount'],
↵errors='coerce')
approved_or_settled['PercentPaid'] = (approved_or_settled['CloseAmount'] /
↵approved_or_settled['ClaimAmount']) * 100

approved_or_settled

average_percent_paid = approved_or_settled['PercentPaid'].mean()

average_percent_paid

```

c:\Users\HOME-PC\anaconda3\lib\site-packages\pandas\core\indexing.py:1773:

SettingWithCopyWarning:

A value is trying to be set on a copy of a slice from a DataFrame.

Try using .loc[row\_indexer,col\_indexer] = value instead

See the caveats in the documentation: [https://pandas.pydata.org/pandas-docs/stable/user\\_guide/indexing.html#returning-a-view-versus-a-copy](https://pandas.pydata.org/pandas-docs/stable/user_guide/indexing.html#returning-a-view-versus-a-copy)

```
self._setitem_single_column(ilocs[0], value, pi)
```

C:\Users\HOME-PC\AppData\Local\Temp\ipykernel\_16928\1115862748.py:8:

SettingWithCopyWarning:

A value is trying to be set on a copy of a slice from a DataFrame.

Try using .loc[row\_indexer,col\_indexer] = value instead

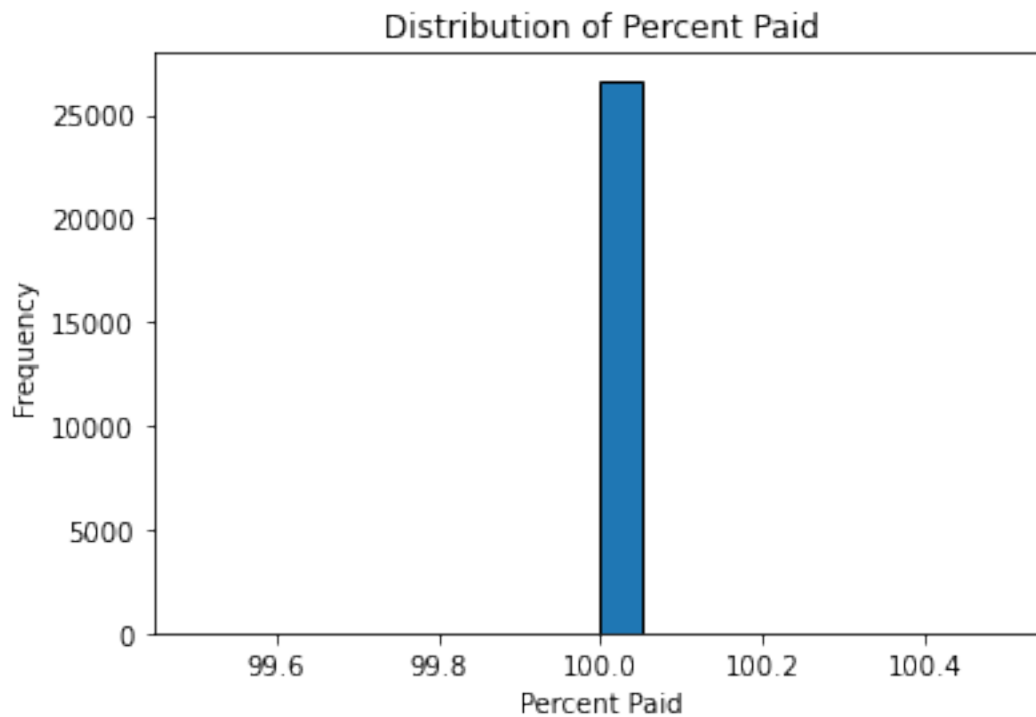
See the caveats in the documentation: [https://pandas.pydata.org/pandas-docs/stable/user\\_guide/indexing.html#returning-a-view-versus-a-copy](https://pandas.pydata.org/pandas-docs/stable/user_guide/indexing.html#returning-a-view-versus-a-copy)

```
approved_or_settled['PercentPaid'] = (approved_or_settled['CloseAmount'] /
approved_or_settled['ClaimAmount']) * 100
```

```
[138]: 100.0
```

```
[139]: import matplotlib.pyplot as plt

plt.hist(approved_or_settled['PercentPaid'], bins=20, edgecolor='black')
plt.title('Distribution of Percent Paid')
plt.xlabel('Percent Paid')
plt.ylabel('Frequency')
plt.show()
```



Q7

```
[85]: airport_claim_counts = df.groupby('AirportName').size()

airport_claim_counts = airport_claim_counts.reset_index(name='ClaimNumber')
```

```
top_airports = airport_claim_counts.sort_values(by='ClaimNumber',
↪ascending=False).head(5)

top_airports
```

```
[85]:
```

|     | AirportName                          | ClaimNumber |
|-----|--------------------------------------|-------------|
| 199 | John F. Kennedy International        | 9232        |
| 251 | Los Angeles International Airport    | 7260        |
| 311 | Newark International Airport         | 6866        |
| 69  | Chicago O'Hare International Airport | 6843        |
| 282 | Miami International Airport          | 6432        |

Q8

```
[92]: newdf.loc[:, 'DateReceived'] = pd.to_datetime(newdf['DateReceived'])

total_close_amount_over_time = newdf.groupby(newdf['DateReceived'].dt.
↪year)['CloseAmount'].sum()

total_close_amount_over_time
```

c:\Users\HOME-PC\anaconda3\lib\site-packages\pandas\core\indexing.py:1773:

SettingWithCopyWarning:

A value is trying to be set on a copy of a slice from a DataFrame.

Try using .loc[row\_indexer,col\_indexer] = value instead

See the caveats in the documentation: [https://pandas.pydata.org/pandas-docs/stable/user\\_guide/indexing.html#returning-a-view-versus-a-copy](https://pandas.pydata.org/pandas-docs/stable/user_guide/indexing.html#returning-a-view-versus-a-copy)

```
self._setitem_single_column(ilocs[0], value, pi)
```

```
[92]: DateReceived
1994                                $50.00
1996                                $0.00
2000                        $0.00 $50.00 $0.00
2002    $350.00 $50.00 $227.92 $50.00 $43.49 $17.00 $1...
2003    $1;000.00 $26.00 $158.00 $10.00 $39.00 $50.00 ...
2004    $0.00 $480.00 $49.95 $0.00 $10.55 $81.25 $120...
2005    $395.00 $0.00 $0.00 $0.00 $0.00 $0.00 $62.11 $...
2006    $49.86 $28.75 $0.00 $0.00 $0.00 $0.00 $0.00 $0...
2007    $162.99 $0.00 $0.00 $0.00 $0.00 $25.00 $206.74...
2008    $189.44 $445.00 $20.00 $192.37 $220.00 $0.00 $...
2009    $0.00 $0.00 $0.00 $0.00 $0.00 $0.00 $0.00 $0.0...
2010                                0
2011                                0
2012                                0
2013                                0
```

|      |          |
|------|----------|
| 2014 | 0        |
| 2015 | 0        |
| 2020 | \$179.08 |
| 2044 | 0        |

Name: CloseAmount, dtype: object

Q9

```
[120]: import matplotlib.pyplot as plt
from shapely.geometry import Point
import geopandas as geo

usa = geo.read_file('maps/states.shp')
usa

ca_map = usa[usa.STATE_ABBR == 'CA']

ca_map.plot(color='white', edgecolor='black')
```

c:\Users\HOME-PC\anaconda3\lib\site-packages\geopandas\plotting.py:48:  
ShapelyDeprecationWarning: The 'type' attribute is deprecated, and will be removed in the future. You can use the 'geom\_type' attribute instead.  
if geom.type.startswith(prefix):

-----  
**TypeError** Traceback (most recent call last)

~\AppData\Local\Temp\ipykernel\_16928\3783271372.py in <module>

```
9
10
---> 11 ca_map.plot(color='white', edgecolor='black')
12
13
```

c:\Users\HOME-PC\anaconda3\lib\site-packages\geopandas\geodataframe.py in

```
↳ plot(self, *args, **kwargs)
897         from there.
898         """
--> 899         return plot_dataframe(self, *args, **kwargs)
900
901         plot.__doc__ = plot_dataframe.__doc__
```

c:\Users\HOME-PC\anaconda3\lib\site-packages\geopandas\plotting.py in

```
↳ plot_dataframe(df, column, cmap, color, ax, cax, categorical, legend, scheme,
↳ k, vmin, vmax, markersize, figsize, legend_kws, categories,
↳ classification_kws, missing_kws, aspect, **style_kws)
613
614     if column is None:
```



```

--> 615         return plot_series(
        616             df.geometry,
        617             cmap=cmap,

c:\Users\HOME-PC\anaconda3\lib\site-packages\geopandas\plotting.py in
↳ plot_series(s, cmap, color, ax, figsize, aspect, **style_kwds)
    411
    412         values_ = values[poly_idx] if cmap else None
--> 413         _plot_polygon_collection(
    414             ax, polys, values_, facecolor=facecolor, cmap=cmap,
↳ **style_kwds
    415         )

c:\Users\HOME-PC\anaconda3\lib\site-packages\geopandas\plotting.py in
↳ _plot_polygon_collection(ax, geoms, values, color, cmap, vmin, vmax, **kwargs)
    127     from matplotlib.collections import PatchCollection
    128
--> 129     geoms, multiindex = _flatten_multi_geoms(geoms)
    130     if values is not None:
    131         values = np.take(values, multiindex, axis=0)

c:\Users\HOME-PC\anaconda3\lib\site-packages\geopandas\plotting.py in
↳ _flatten_multi_geoms(geoms, prefix)
    47     for ix, geom in enumerate(geoms):
    48         if geom.type.startswith(prefix):
---> 49             for poly in geom:
    50                 components.append(poly)
    51                 component_index.append(ix)

TypeError: 'MultiPolygon' object is not iterable

```

